



Precision Medicine: A Machine  
Learning Approach to Diabetes  
Classification  
Diabetes Classifier Application  
User Guide

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## 1. Introduction

Welcome to the Diabetes Classification dashboard. This tool will allow users to view different data visualizations, as well as three different machine learning classification models used to classify patients with or without diabetes.

**Version:** 3.1

**Release Date:** September 30, 2024

**Support Contact:** [ewagner@my.gcu.edu](mailto:ewagner@my.gcu.edu)

## Getting Started

### 2.1 System Requirements

- Browser: Latest versions of Safari, Google Chrome, Microsoft Edge, or Firefox
- Operating System: Current versions of Windows or macOS
- Internet Connection: Stable connection

### 2.2 Starting the API

- Open the web browser and go to the Diabetes Classification API URL:  
<https://diabetesclassifierapplicationsanmarco.streamlit.app/>
- The API will load with the with home page:

#### Diabetes Classifier Application

This application aims to provide insight to the Diabetes Health Indicators Dataset through data exploration. Additionally this application showcases three classification models used in conjunction with the data set.

[Home](#) [Exploratory Data Analysis](#) [K-Nearest Neighbors Classifier](#) [Random Forest Classifier](#) [MLP Classifier](#)

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Rating and Feedback

Please provide a rating and feedback to let us know how we can improve the application for future users

Feedback:

[Submit](#)

## Dashboard Overview

## 2.2.1 Site Tabs

# Diabetes Classifier Application

This application aims to provide insight to the Diabetes Health Indicators Dataset through data explo

[Home](#) [Exploratory Data Analysis](#) [K-Nearest Neighbors Classifier](#) [Random Forest Classifier](#) [MLP Classifier](#)

The API is designed with organization in mind, each section of the API is broken down into tabs. Fin explanations of each tab below:

### 1) Home Tab

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Rating and Feedback

Please provide a rating and feedback to let us know how we can improve the application for future users

Feedback:

[Submit](#)

- Here users will find the initial help button, as well as a feature to rate the GUI and leave any feedback for the developer.
- Upon clicking the [CLICK ME FOR HELP](#) tab a brief paragraph summarizing the components of the API is displayed.

[CLICK ME FOR HELP](#)

The Diabetes Health Indicators Dataset is preloaded into the application. Users may explore the application thorough clicking the various tabs. In the Exploratory Data Analysis tab the user will find data exploration componentes such as the dataframe, insightful graphs and visuals, and relative information about the data such as missing values and data types. Each model has a tab dedicated to run that model and present its performance metrics. For example the K-Nearest Neighbors tab will allow users to run the model by simply clicking the button. Upon click the model will run and the results will populate within the tab.

## 2) Exploratory Data Analysis Tab

Home [Exploratory Data Analysis](#) K-Nearest Neighbors Classifier Random Forest Classifier MLP Classifier

### Diabetes Health Indicators Data Set

|   | Diabetes_012 | HighBP | HighChol | CholCheck | BMI | Smoker | Stroke | HeartDiseaseorAttack | PhysActivity | Fruits | Veggies | HvyAlcoholConsump | AnyHealthcare | NoDocbcCost | GenHlth | MentHlth | PhysHlth | DiffWalk | Sex | Age | Education | Income |
|---|--------------|--------|----------|-----------|-----|--------|--------|----------------------|--------------|--------|---------|-------------------|---------------|-------------|---------|----------|----------|----------|-----|-----|-----------|--------|
| 0 | 0            | 1      | 1        | 1         | 40  | 1      | 0      | 0                    | 0            | 0      | 1       | 0                 | 1             | 0           | 5       | 18       | 15       | 1        | 0   | 9   | 4         | 3      |
| 1 | 0            | 0      | 0        | 0         | 25  | 1      | 0      | 0                    | 1            | 0      | 0       | 0                 | 0             | 1           | 3       | 0        | 0        | 0        | 0   | 7   | 6         | 1      |
| 2 | 0            | 1      | 1        | 1         | 28  | 0      | 0      | 0                    | 0            | 1      | 0       | 0                 | 1             | 1           | 5       | 30       | 30       | 1        | 0   | 9   | 4         | 8      |
| 3 | 0            | 1      | 0        | 1         | 27  | 0      | 0      | 0                    | 1            | 1      | 1       | 0                 | 1             | 0           | 2       | 0        | 0        | 0        | 0   | 11  | 3         | 6      |
| 4 | 0            | 1      | 1        | 1         | 24  | 0      | 0      | 0                    | 1            | 1      | 1       | 0                 | 1             | 0           | 2       | 3        | 0        | 0        | 0   | 11  | 5         | 4      |
| 5 | 0            | 1      | 1        | 1         | 25  | 1      | 0      | 0                    | 1            | 1      | 1       | 0                 | 1             | 0           | 2       | 0        | 2        | 0        | 1   | 10  | 6         | 8      |
| 6 | 0            | 1      | 0        | 1         | 30  | 1      | 0      | 0                    | 0            | 0      | 0       | 0                 | 1             | 0           | 3       | 0        | 14       | 0        | 0   | 9   | 6         | 7      |

- This tab contains the Diabetes Health Indicators Dataset, a section with interactive buttons, interactive dropdowns, interactive dropdowns with visuals, and the final cleaned data set.

Diabetes Health Indicators Data Set Interactive Buttons

Data Shape

Null values within the Diabetes Health Indicators Dataset

Descriptive Statistics of original Diabetes Health Indicators Dataset

Number of duplicate values in the original Diabetes Health Indicators Dataset

Number of duplicate values in the cleaned Diabetes Health Indicators Dataset

Diabetes Health Indicators Data Set Interactive Drop-downs

Cleaned column names

Variable Data Types

Description of the variables found in the Diabetes Health Indicators dataset

Diabetes Class counts

Resampled Diabetes Class counts

Interactive Drop-Downs with visuals

Box plots of all features of the Diabetes Health Indicators Dataset

Boxplots with extreme outliers removed

Original Class Distribution

Class Distribution after Random Over-Sampling

Feature Selection via Correlation Matrix

Bar chart showing correlation between Diabetes Class and other variables(easier comprehension for non technical users)

Selected features and their correlation

Bar chart showing correlation between Diabetes Class and selected variables(easier comprehension for non technical users)

Bar charts used to show the density of important selected variables

Final Diabetes Health Indicators Dataframe

Cleaned, Balanced Data with feature selection applied

For security purposes all identifying patient information has been removed

|    | Diabetes_Class | High_BP | High_Chol | BMI | Smoker | Stroke | Heart_Disease_or_Attack | Phys_Activity_In_Last_30_Days | Hvy_Alcohol_Consump | Gen_Hlth | Ment_Hlth | Phys_Hlth | Diff_Walk | Sex | Age |
|----|----------------|---------|-----------|-----|--------|--------|-------------------------|-------------------------------|---------------------|----------|-----------|-----------|-----------|-----|-----|
| 1  | 0              | 0       | 0         | 25  | 1      | 0      | 0                       | 1                             | 0                   | 3        | 0         | 0         | 0         | 0   | 7   |
| 3  | 0              | 1       | 0         | 27  | 0      | 0      | 0                       | 1                             | 0                   | 2        | 0         | 0         | 0         | 0   | 11  |
| 4  | 0              | 1       | 1         | 24  | 0      | 0      | 0                       | 1                             | 0                   | 2        | 3         | 0         | 0         | 0   | 11  |
| 5  | 0              | 1       | 1         | 25  | 1      | 0      | 0                       | 1                             | 0                   | 2        | 0         | 2         | 0         | 1   | 10  |
| 7  | 0              | 1       | 1         | 25  | 1      | 0      | 0                       | 1                             | 0                   | 3        | 0         | 0         | 1         | 0   | 11  |
| 9  | 0              | 0       | 0         | 24  | 0      | 0      | 0                       | 0                             | 0                   | 2        | 0         | 0         | 0         | 1   | 8   |
| 10 | 2              | 0       | 0         | 25  | 1      | 0      | 0                       | 1                             | 0                   | 3        | 0         | 0         | 0         | 1   | 13  |
| 13 | 2              | 1       | 1         | 28  | 0      | 0      | 0                       | 0                             | 0                   | 4        | 0         | 0         | 1         | 0   | 11  |
| 15 | 0              | 1       | 0         | 33  | 0      | 0      | 0                       | 1                             | 0                   | 2        | 5         | 0         | 0         | 0   | 6   |
| 16 | 0              | 1       | 1         | 21  | 0      | 0      | 0                       | 1                             | 0                   | 3        | 0         | 0         | 0         | 0   | 10  |

i) The buttons on this page are:

- Data Shape

- Null values within the Diabetes Health Indicators Dataset
- Descriptive Statistics of the original Diabetes Health Indicators Dataset
- Number of Duplicated values in the original Diabetes Health Indicators Dataset
- Number of duplicate values in the cleaned Diabetes Health Indicators Dataset

ii) The Dropdowns on this page are:

- Cleaned column names
- Variable Data Types
- Description of the variables found in the Diabetes Health Indicators dataset
- Diabetes Class counts
- Resampled Diabetes Class counts

Interactive Dropdowns with visuals

- Box plots of all features of the Diabetes Health Indicators Dataset
- Boxplots with extreme outliers removed
- Original Class Distribution
- Class Distribution after Random Over-Sampling
- Feature Selection via Correlation Matrix
- Bar chart showing correlation between Diabetes Class and other variables (easier comprehension for non-technical users)
- Selected features and their correlation

- Bar chart showing correlation between Diabetes Class and selected variables (easier comprehension for non-technical users)
- Bar charts used to show the density of important selected variables

## 2) K-Nearest Neighbors Classifier Tab

Home Exploratory Data Analysis **K-Nearest Neighbors Classifier**

Run KNN Classifier

This tab is for the user to run the KNN Classifier and view its results  
There is one button on the page: Run KNN Classifier

## 3) Random Forest Classifier

Home Exploratory Data Analysis K-Nearest Neighbors Classifier **Random Forest Classifier**

Run Random Forest Classifier

This tab is for the user to run the Random Forest Classifier and view its results  
There is one button on the page: Run Random Forest Classifier

## 4) MLP Classifier Tab

Home Exploratory Data Analysis K-Nearest Neighbors Classifier Random Forest Classifier **MLP Classifier**

Run MLP Classifier

This tab is for the user to run the MLP Classifier and view its results  
There is one button on the page: Run MLP Classifier

## Using the Home Tab Dashboard

The Home tab is used to inform users about the application and provide feedback

- 1) Click the “Click MR FOR HELP” button

(a) Upon clicking the button text will appear explain how to use the application

CLICK ME FOR HELP

The Diabetes Health Indicators Dataset is preloaded into the application. Users may explore the application through clicking the various tabs. In the Exploratory Data Analysis tab the user will find data exploration components such as the dataframe, insightful graphs and visuals, and relative information about the data such as missing values and data types. Each model has a tab dedicated to run that model and present its performance metrics. For example the K-Nearest Neighbors tab will allow users to run the model by simply clicking the button. Upon click the model will run and the results will populate within the tab.

2. Using the stars the user may select a rating for the application

- Clicking a star will cause it to light up yellow; the more stars are highlighted the better the rating



Rating and Feedback

3. The feedback box is where users type any feedback for the application developer

- Click the gray box within the feedback box to type the feedback
- Once ready click the submit button to send the feedback

Please provide a rating and feedback to let us know how we can improve the application for future users

Feedback:

Submit

## Using the Exploratory Data Analysis Tab

- The Exploratory Data Analysis Tab will load presenting the original unaltered data set.
- Below this dataset the application is broken into two columns with buttons on the left and dropdowns on the right

## Diabetes Health Indicators Data Set Interactive Buttons:

- Upon clicking on the Data shape button, the shape of the data is displayed:

Data Shape

(253680, 22)

- Upon clicking on the “Null Values within the Diabetes Health Indicators Dataset: button, null values are displayed in a table:

Null Values within the Diabetes Health Indicators Dataset

|                      | 0 |
|----------------------|---|
| Diabetes_012         | 0 |
| HighBP               | 0 |
| HighChol             | 0 |
| CholCheck            | 0 |
| BMI                  | 0 |
| Smoker               | 0 |
| Stroke               | 0 |
| HeartDiseaseorAttack | 0 |
| PhysActivity         | 0 |
| Fruits               | 0 |
|                      |   |

- Upon clicking on the “Descriptive Statistics of the original Diabetes Health Indicators Dataset” button, the application will run a code to display the descriptive

Descriptive Statistics of original Diabetes  
Health Indicators Dataset

|                      | count   | mean    | std    |
|----------------------|---------|---------|--------|
| Diabetes_012         | 253,680 | 0.2969  | 0.6982 |
| HighBP               | 253,680 | 0.429   | 0.4949 |
| HighChol             | 253,680 | 0.4241  | 0.4942 |
| CholCheck            | 253,680 | 0.9627  | 0.1896 |
| BMI                  | 253,680 | 28.3824 | 6.6087 |
| Smoker               | 253,680 | 0.4432  | 0.4968 |
| Stroke               | 253,680 | 0.0406  | 0.1973 |
| HeartDiseaseorAttack | 253,680 | 0.0942  | 0.2931 |

statistics in a table:

- Upon clicking on the “Number of duplicate Values in the original Diabetes Health Indicators Dataset” button, the application will display the number of duplicate entries:

Number of duplicate values in the original  
Diabetes Health Indicators Dataset

23899

- Upon clicking on the “Number of duplicate Values in the cleaned Diabetes Health Indicators Dataset” button, the application will display the number of duplicate entries:

Number of duplicate values in the cleaned  
Diabetes Health Indicators Dataset

0

## Diabetes Health Indicators Data Set Interactive Dropdowns

Upon toggle of the “Cleaned column names” expander the cleaned column names are displayed:

# Cleaned column names

Column names were cleaned for easier comprehension

|   |                              |
|---|------------------------------|
|   | 0                            |
| 0 | Diabetes_Class               |
| 1 | High_BP                      |
| 2 | High_Chol                    |
| 3 | Had_Chol_Check_In_Last_5_Yea |
| 4 | BMI                          |
| 5 | Smoker                       |
| 6 | Stroke                       |

Upon toggle of the “Variable Data Types” expander the transformed variable type is displayed:

| Variable Data Types           |       | ^ |  |
|-------------------------------|-------|---|--|
|                               | 0     |   |  |
| Diabetes_Class                | int64 |   |  |
| High_BP                       | int64 |   |  |
| High_Chol                     | int64 |   |  |
| Had_Chol_Check_In_Last_5_Yea  | int64 |   |  |
| BMI                           | int64 |   |  |
| Smoker                        | int64 |   |  |
| Stroke                        | int64 |   |  |
| Heart_Disease_or_Attack       | int64 |   |  |
| Phys_Activity_In_Last_30_Days | int64 |   |  |
| Consumes_Fruits_Daily         | int64 |   |  |

Upon toggle of the “Description of the variables found in the Diabetes Health Indicators dataset” expander a description of the features is displayed:

Description of the variables found in the Diabetes Health Indicators dataset ^

Description of features for easier comprehension of variables

|                | Description            |
|----------------|------------------------|
| Diabetes_Class | Diabetes status        |
| HighBP         | High blood pressure    |
| HighChol       | High cholesterol: (>24 |
| CholCheck      | Checked cholesterol i  |
| BMI            | Body mass index        |
| Smoker         | Smoked at least 100 c  |
| Stroke         | Had stroke or told by  |

Upon toggle of the “Diabetes Class counts” expander the original class counts are displayed:

### Diabetes Class counts

| Diabetes_Class | proportion |
|----------------|------------|
| 0              | 0.8271     |
| 2              | 0.1527     |
| 1              | 0.0201     |

Upon toggle of the “Resampled Diabetes Class counts” expander the resampled class

Resampled Diabetes Class counts

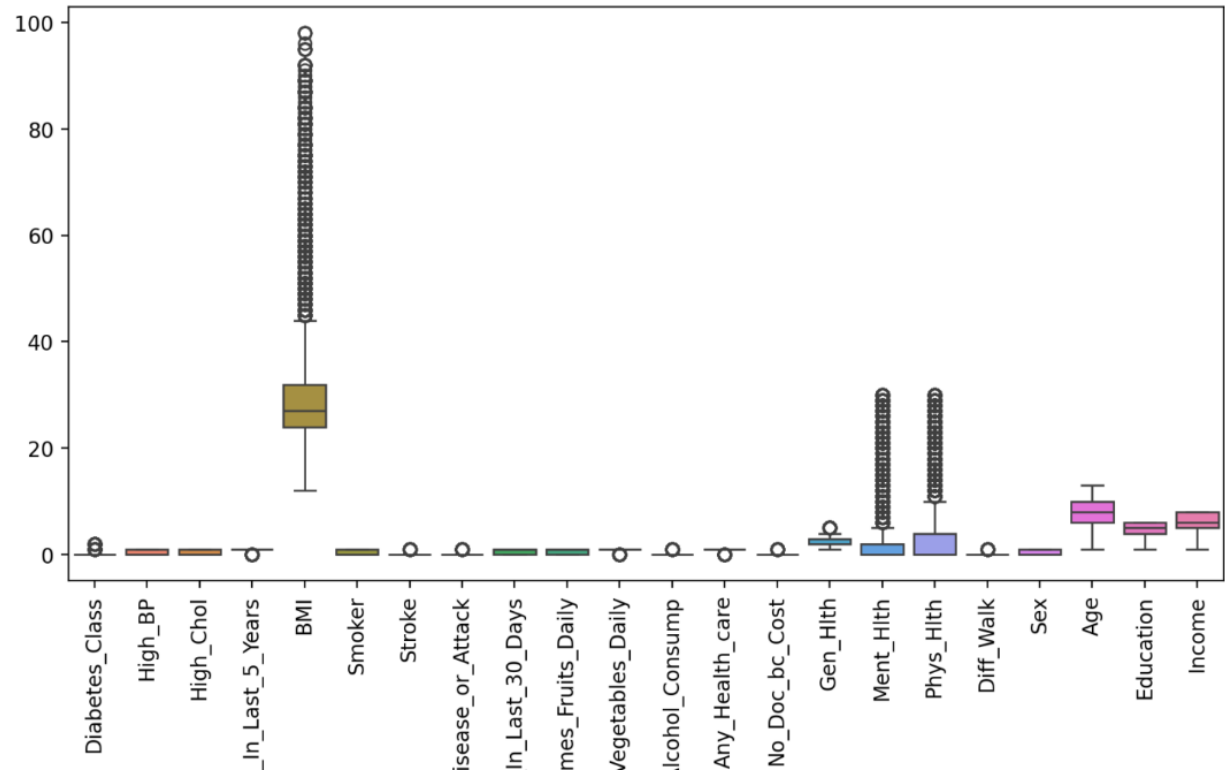
| Diabetes_Class | proportion |
|----------------|------------|
| 0              | 0.3333     |
| 2              | 0.3333     |
| 1              | 0.3333     |

counts are displayed:

Interactive Dropdowns with visuals

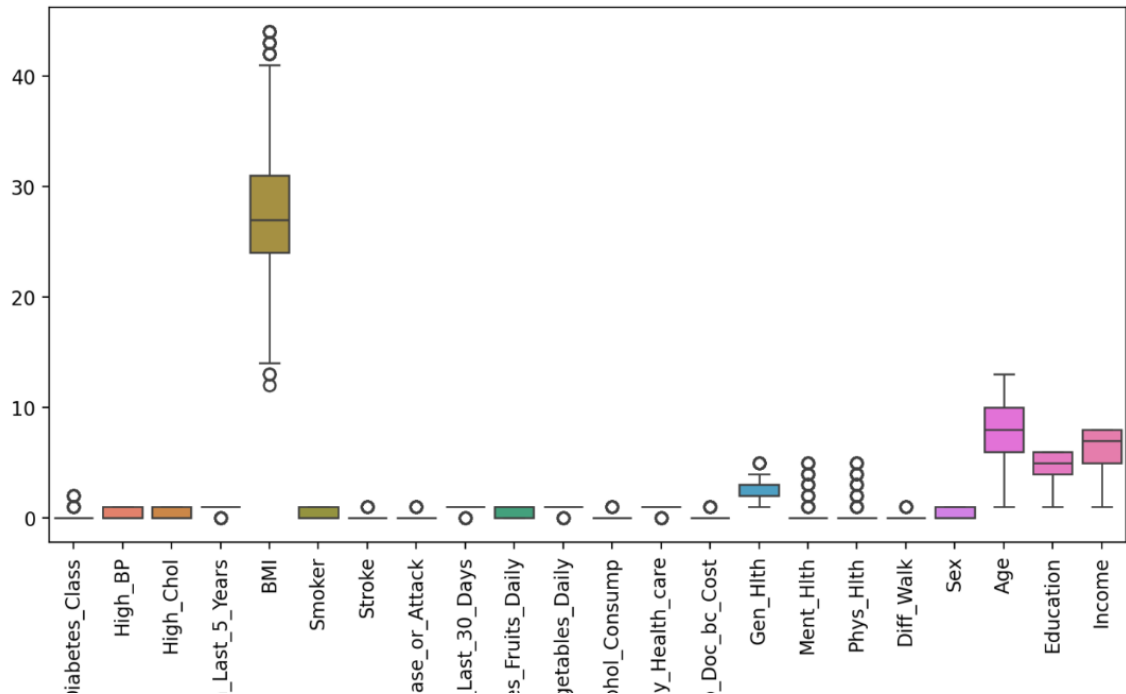
Upon toggle of the “Box plots of all features of the Diabetes Health Indicators Dataset” box plots are displayed for each variable:

Box plots of all features of the Diabetes Health Indicators Dataset



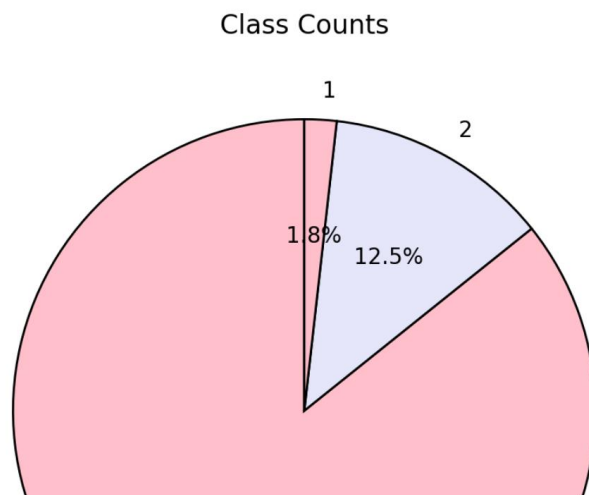
Upon toggle of the “Boxplots with extreme outliers removed” box plots are displayed for each variable with the outliers removed:

Boxplots with extreme outliers removed

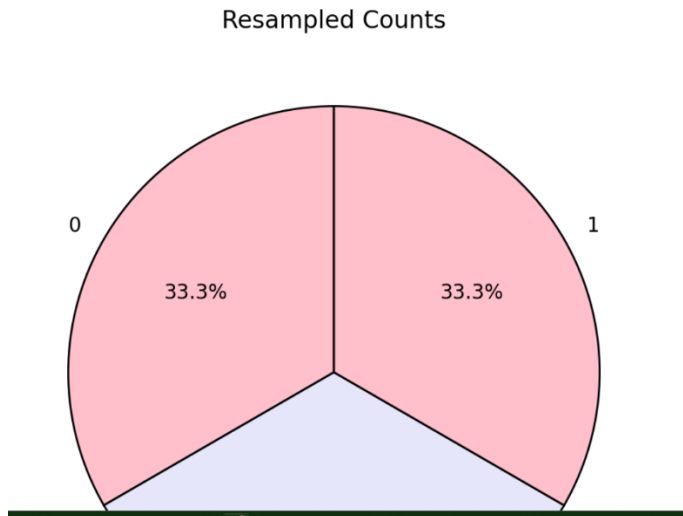


Upon toggle of the “Original Class Distribution” pie charts are displayed showing the original class distribution:

Distribution of classes before over sampling is used

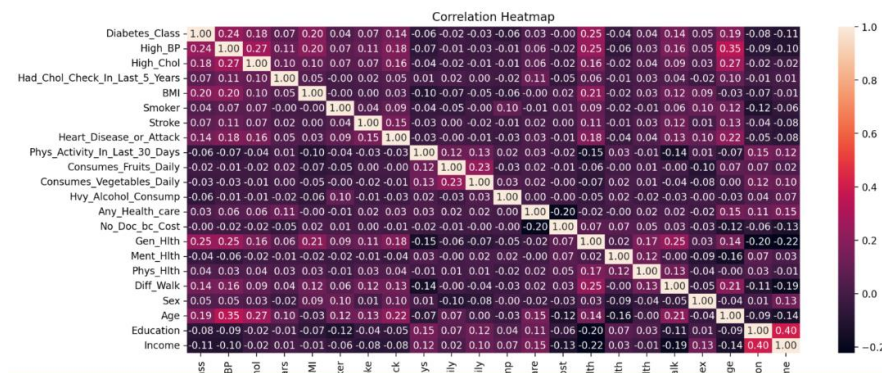


Upon toggle of the “Original Class Distribution” pie charts are displayed showing the original class distribution:

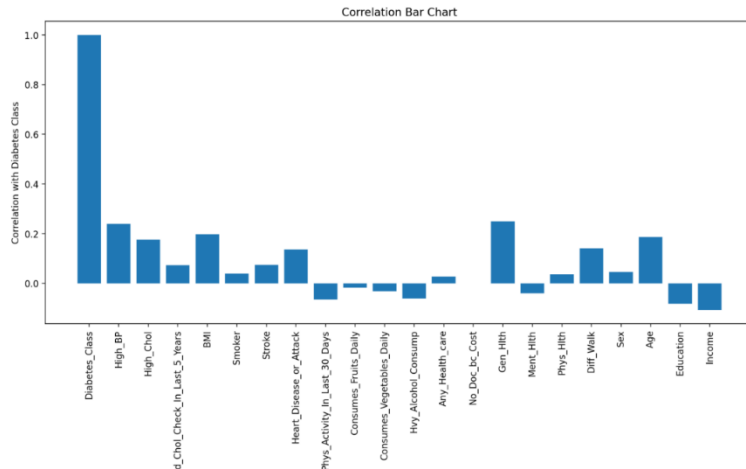


Upon toggle of the “Feature Selection via Correlation Matrix” expander a heat map is displayed showing the correlation of the variables:

Correlation matrix used to determine feature selection. Features are selected at with a value of .2 or greater, unless the variable was noted as an indicator by the WHO, CDC, or American Diabetes Association



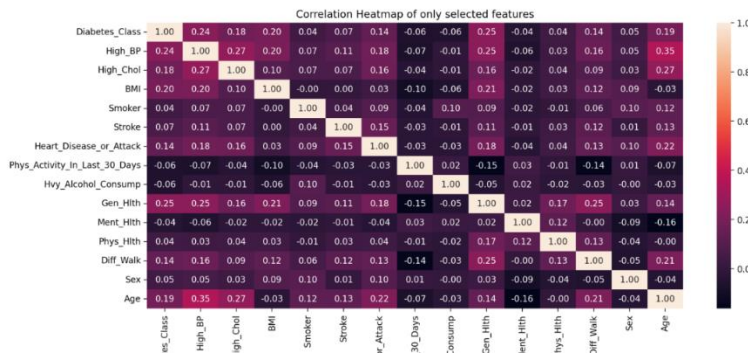
Upon toggle of the “Bar chart showing correlation between Diabetes Class and other variables (easier comprehension for non-technical users)” expander a bar chart is displayed showing the correlation of the variables:



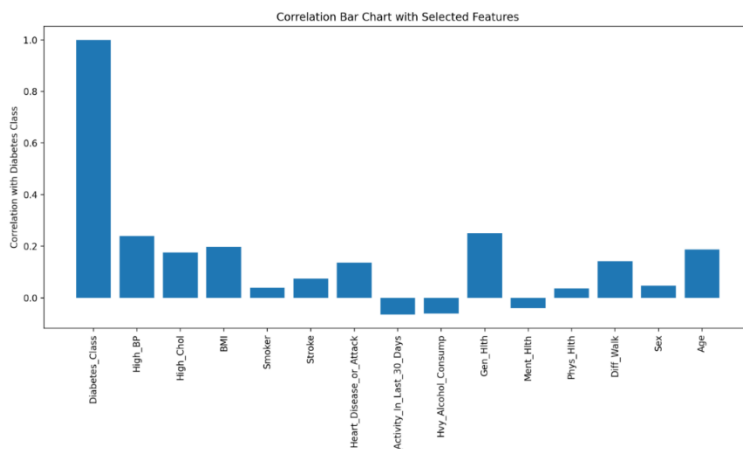
Upon toggle of the “Selected features and their correlation” expander a heat map is displayed showing the correlation of the variables:

Selected features and their correlation

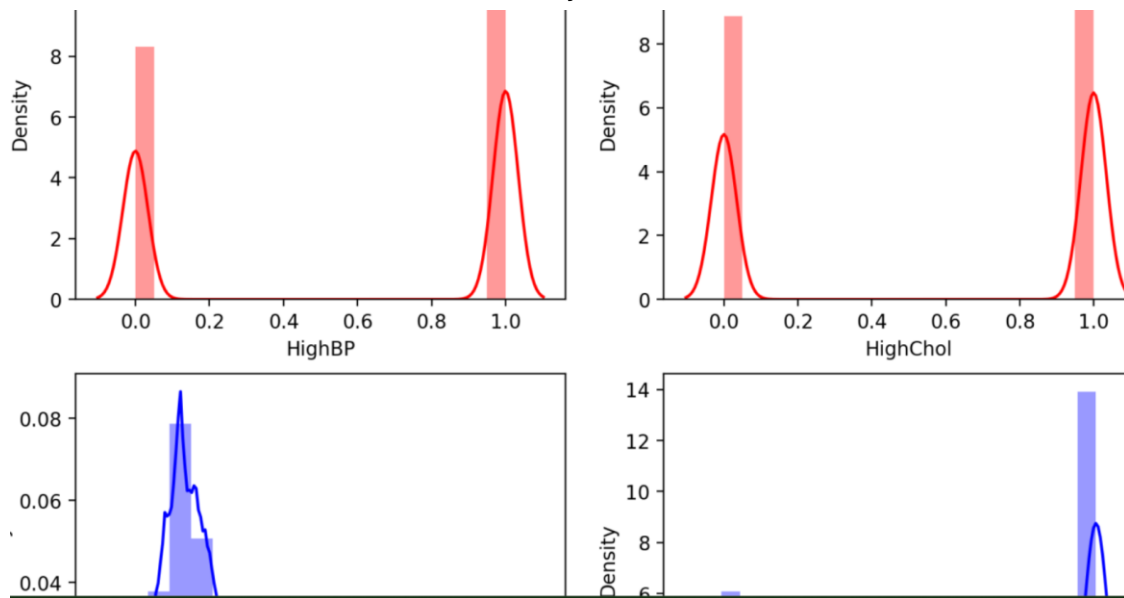
Correlation matrix to show the correlation of the selected features



Upon toggle of the “Bar chart showing correlation between Diabetes Class and selected variables (easier comprehension for non-technical users)” expander a bar chart is displayed showing the correlation of the selected variables:



Upon toggle of the “Bar charts used to show the density of important selected variables” expander bar charts are used to show the density of certain selected features:



## Final Diabetes Health Indicators Dataframe

Upon toggle of the “Final Diabetes Health Indicators Data frame” expander the final clean data frame with feature selection applied is presented:

Cleaned, Balanced Data with feature selection applied

For security purposes all identifying patient information has been removed

|    | Diabetes_Class | High_BP | High_Chol | BMI | Smoker | Stroke | Heart_Disease_or_Attack | P |
|----|----------------|---------|-----------|-----|--------|--------|-------------------------|---|
| 1  | 0              | 0       | 0         | 25  | 1      | 0      | 0                       |   |
| 3  | 0              | 1       | 0         | 27  | 0      | 0      | 0                       |   |
| 4  | 0              | 1       | 1         | 24  | 0      | 0      | 0                       |   |
| 5  | 0              | 1       | 1         | 25  | 1      | 0      | 0                       |   |
| 7  | 0              | 1       | 1         | 25  | 1      | 0      | 0                       |   |
| 9  | 0              | 0       | 0         | 24  | 0      | 0      | 0                       |   |
| 10 | 2              | 0       | 0         | 25  | 1      | 0      | 0                       |   |

## Using the K-Nearest Neighbors Classifier Tab

1. Upon clicking of the “Run KNN Classifier” button the application will run the KNN model and present the performance metrics:

- Accuracy
- Precision

- iii. Recall
- iv. F-1 Score
- v. Confusion Matrix

## KNN Performance Metrics

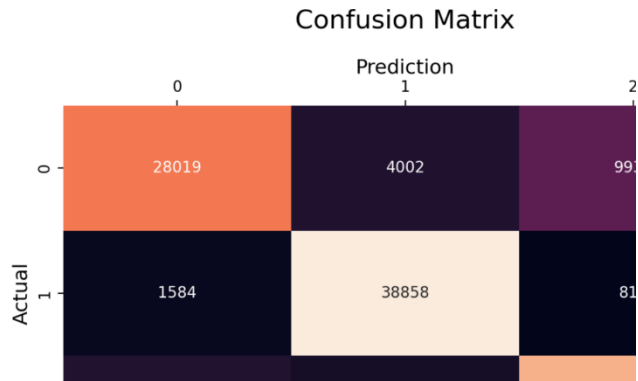
F1\_Score:

| value  |
|--------|
| 0.7402 |
| 0.8927 |
| 0.7881 |

Accuracy KNN: 0.8102994419380705

Precision KNN: 0.8102994419380705

Recall KNN: 0.8102994419380705



## Using the Random Forest Classifier Tab

1. Upon clicking of the “Run Random Forest Classifier” button the application will run the Random Forest model and present the performance metrics:

- i. Accuracy
- ii. Precision
- iii. Recall
- iv. F-1 Score
- v. Confusion Matrix

## Random Forest Performance Metrics

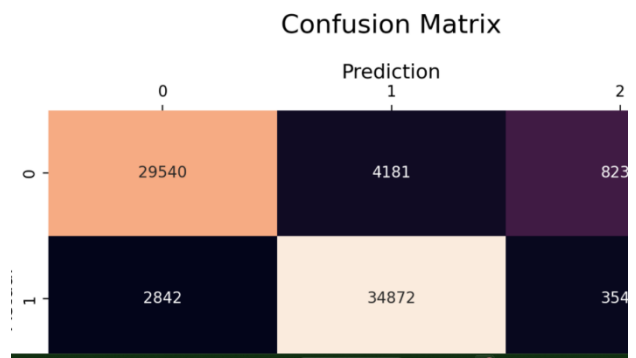
F1\_Score:

| value  |
|--------|
| 0.7472 |
| 0.8204 |
| 0.744  |

Accuracy KNN: 0.7713323019852102

Precision KNN: 0.7713323019852102

Recall KNN: 0.7713323019852102



## Using the MLP Classifier Tab

1. Upon clicking of the “Run MLP Classifier” button the application will run the MLP Classifier model and present the performance metrics:

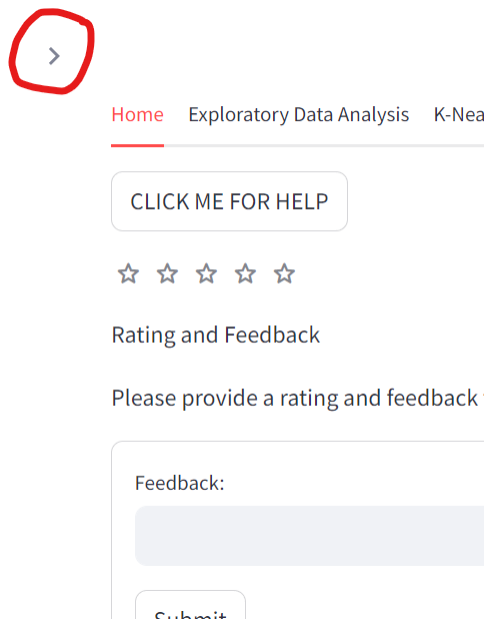
- i. Accuracy
- ii. Precision
- iii. Recall
- iv. F-1 Score
- v. Confusion Matrix

## Validation

### Using the Diabetes Classification Input tool

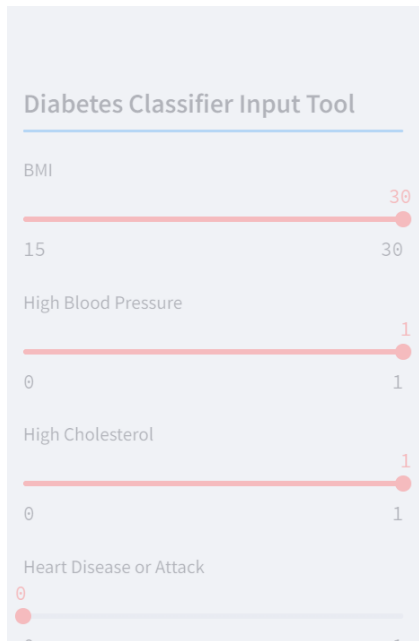
The sidebar allows users to input certain criteria then run the model to make a classification based on the input information.

1. Upon clicking the arrow located on the top left side of the screen a side bar opens



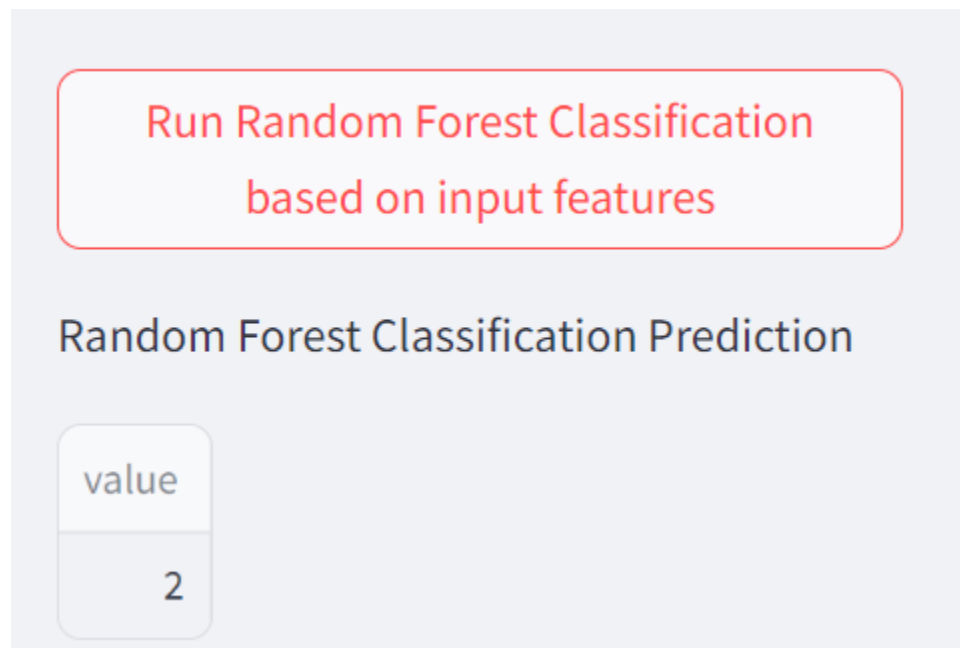
The image shows the "Diabetes Classifier Input Tool" form. It has a title "Diabetes Classifier Input Tool" with a blue underline. Below the title are five input fields, each with a slider bar and a red circle indicating the current value. The fields are: BMI (range 15 to 30, value 30), High Blood Pressure (range 0 to 1, value 0), High Cholesterol (range 0 to 1, value 0), Heart Disease or Attack (range 0 to 1, value 0), and Physical Activity in the Last 30 Days (range 0 to 1, value 0).

2. Upon clicking the circle located on the slider bar the user will drag the orange circle to the desired value



The image shows a web interface titled "Diabetes Classifier Input Tool". It contains four horizontal sliders for input features: BMI (range 15 to 30, set at 30), High Blood Pressure (range 0 to 1, set at 1), High Cholesterol (range 0 to 1, set at 1), and Heart Disease or Attack (range 0 to 1, set at 0). Each slider has a red dot indicating the current value.

3. In the side bar there are three buttons “Run Random Forest Classification based on input features”, “Run KNN Classification based on input features”, and “Run MLP Classification based on input features”
  - iii. Upon clicking of the “Run Random Forest Classification based on input features” the model is run, and a classification is returned.



The image shows a web interface titled "Random Forest Classification Prediction". It features a large red button with the text "Run Random Forest Classification based on input features". Below the button, there is a text input field labeled "value" containing the number "2".

- iv. Upon clicking of the “Run KNN Classification based on input features” the model is run, and a classification is returned.

Run KNN Classification based on input features

KNN Classification Predication

value

2

- v. Upon clicking of the “Run MLP Classification based on input features” the model is run, and a classification is returned.

Run MLP Classification based on input features

MLP Classification Prediction

value

2

### Frequently Asked Questions

1. Is the application able to show predictions given input data?
  - a. Yes! By using the sidebar, the user can select certain input information and run the classification to get an output class
2. Can an updated data set be uploaded to the model?
  - a. No, the current application is set to work and make classifications based on the 2015 BRFSS survey.

## Help and Contact Details

- For help and other information regarding the Code or API please contact Emily Wagner at [ewagner@my.gcu.edu](mailto:ewagner@my.gcu.edu)